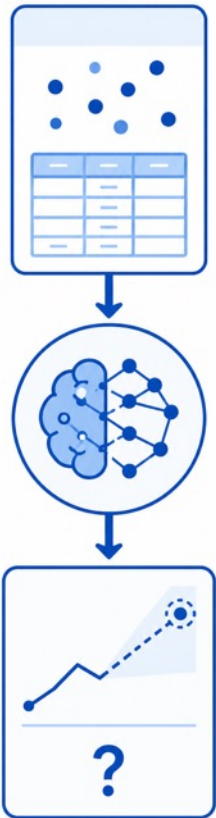


Introduction to Generative AI

Matt Engelhard

Duke Biostatistics & Bioinformatics

Predictive AI versus Generative AI (GenAI)



Predictive AI

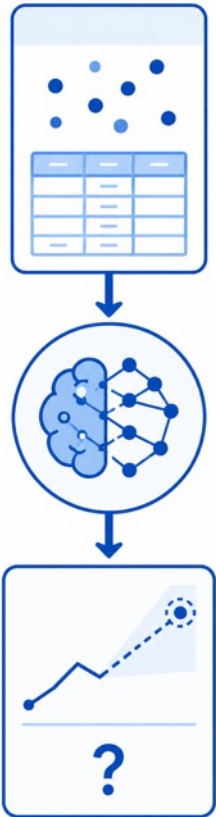
Tasks

- Regression
- Classification
- Risk prediction

Types of data

- Tabular
- Images
- Text
- Biomedical signals
- Multimodal

Predictive AI versus Generative AI (GenAI)



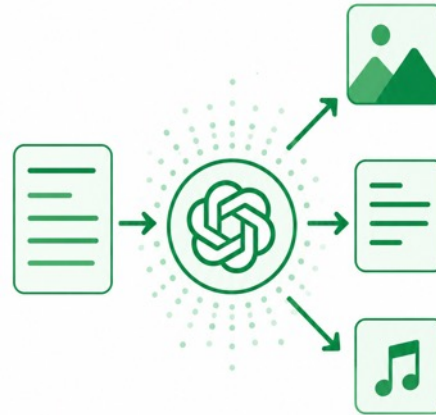
Predictive AI

Tasks

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Types of data

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Generative AI

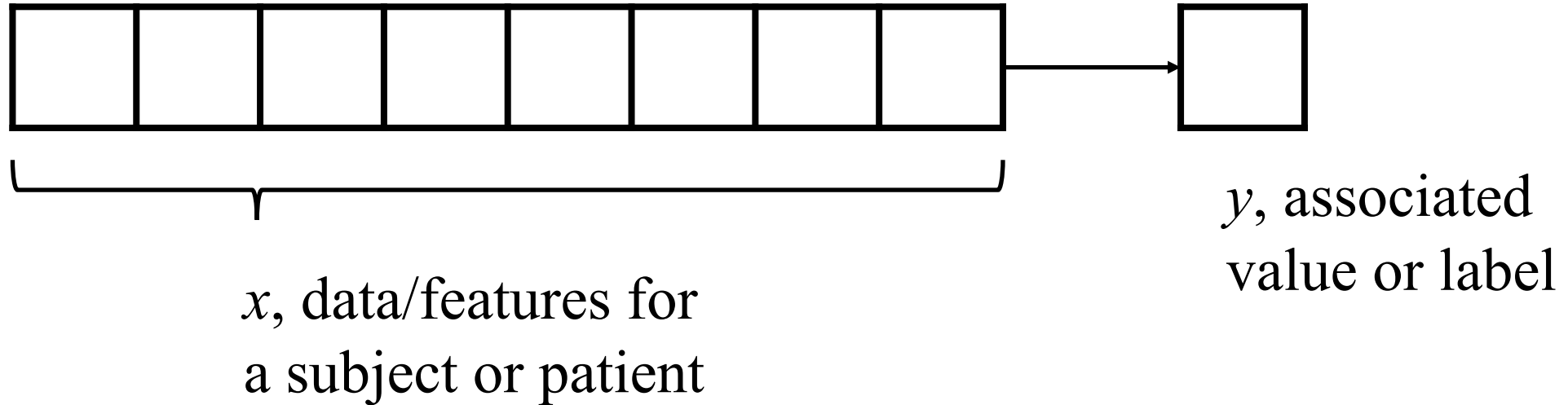
Tasks

- “Data generation”
- Less well-defined
- Can perform pred tasks

Types of data

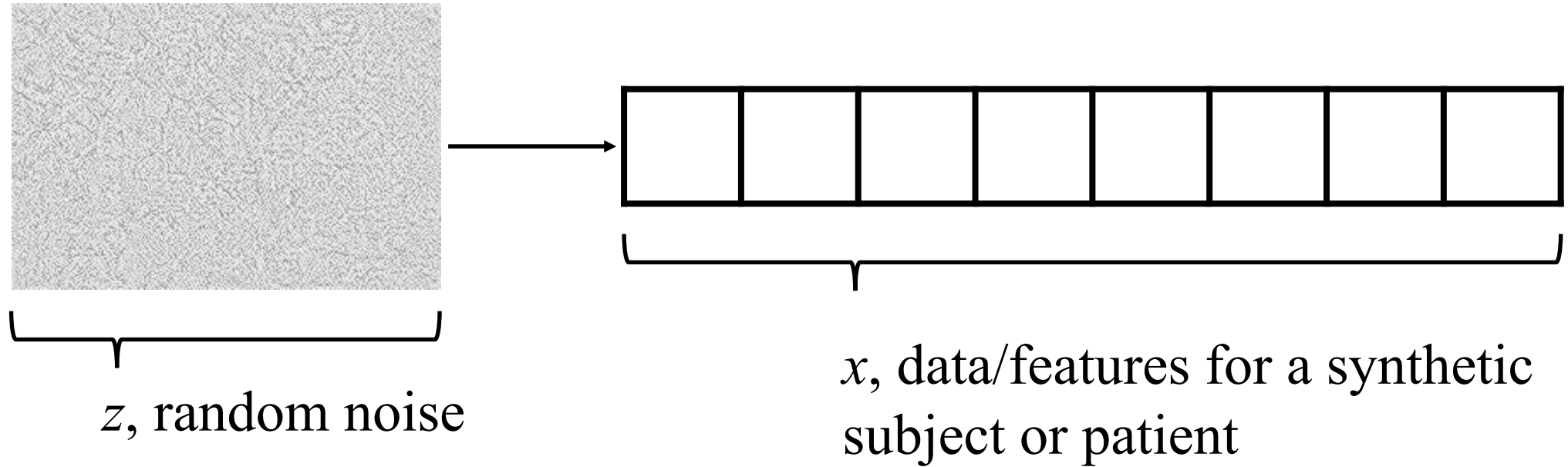
- Tabular
- Images
- **Text**
- Biomedical signals
- **Multimodal**

The Predictive AI Paradigm



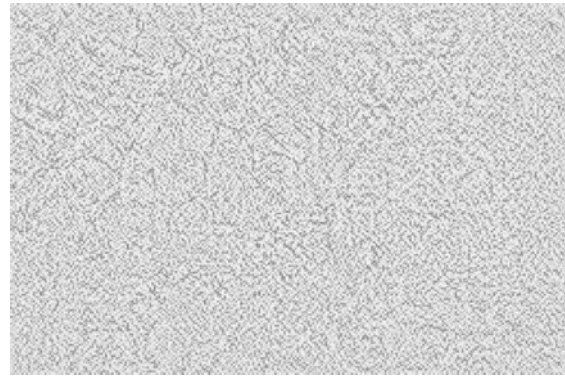
End goal: predict y from x

The Generative AI Paradigm



End goal: create structure from randomness

Image generation via GAN or diffusion



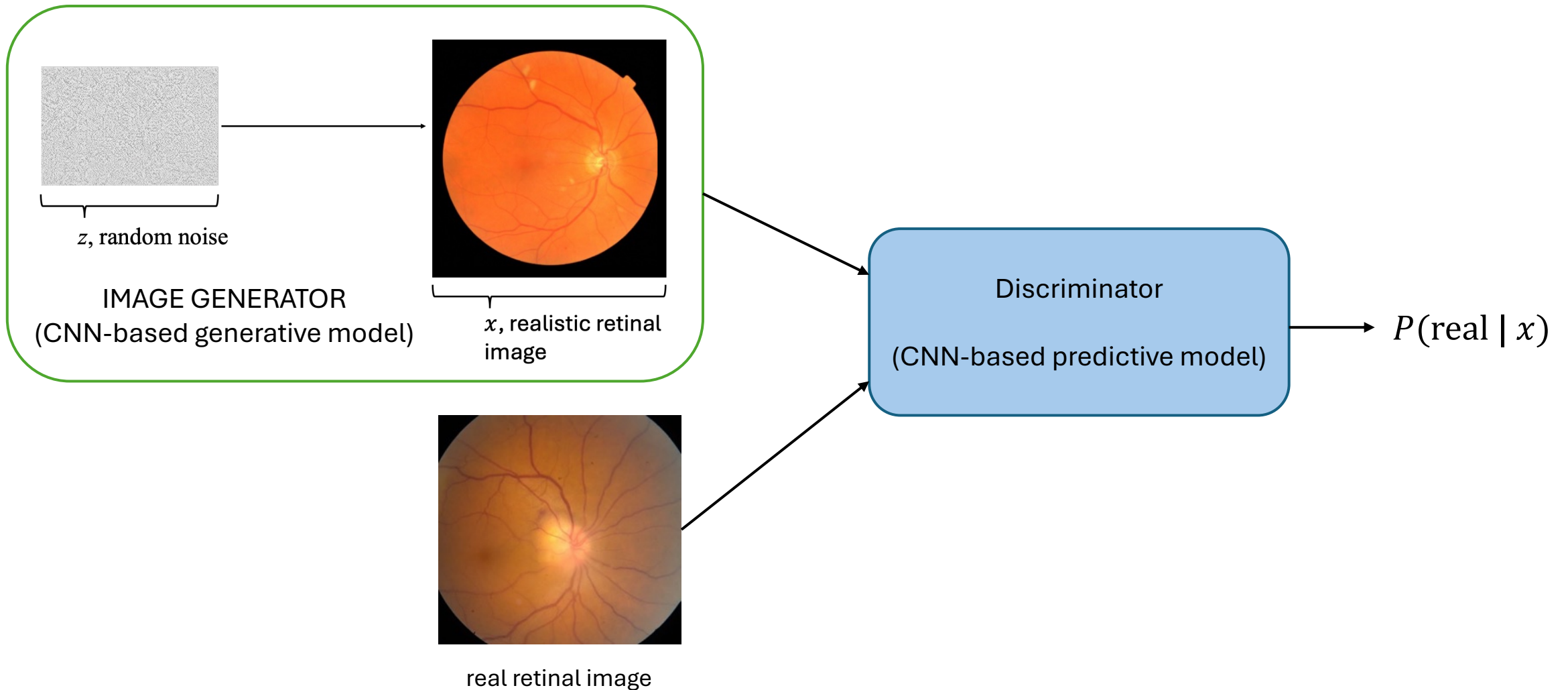
z , random noise



x , realistic retinal image

End goal: create structure from randomness

Generative Adversarial Network (GAN)



Generation via Diffusion

- The model learns to predict what noise was added
- During generation, it progressively de-noises
- For images, the diffusion process is typically performed on latent image representations

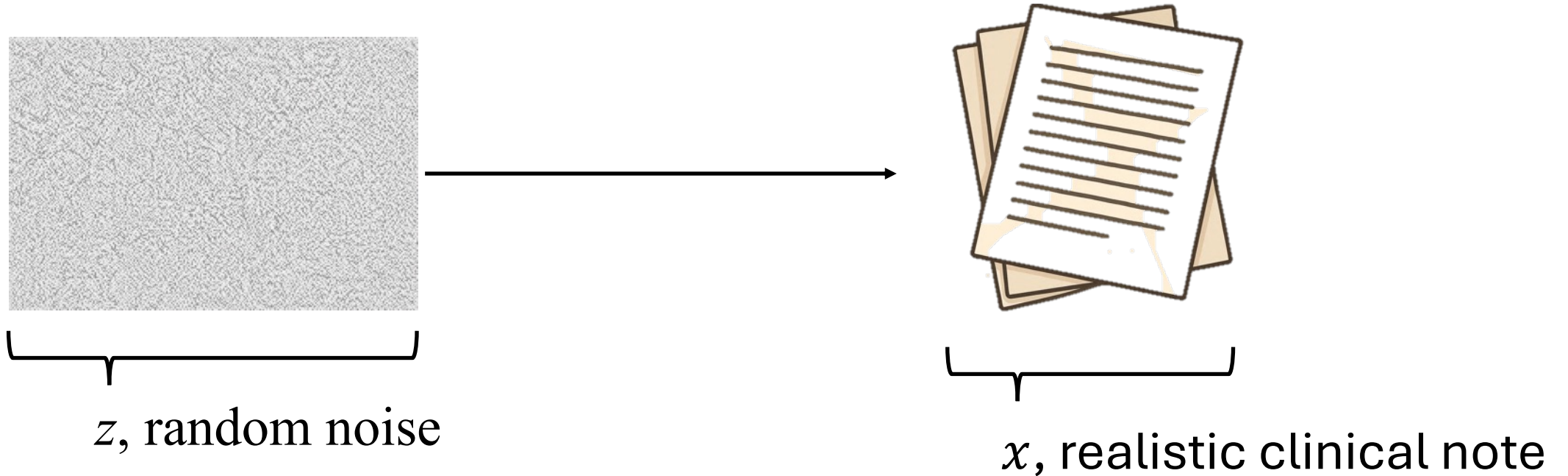


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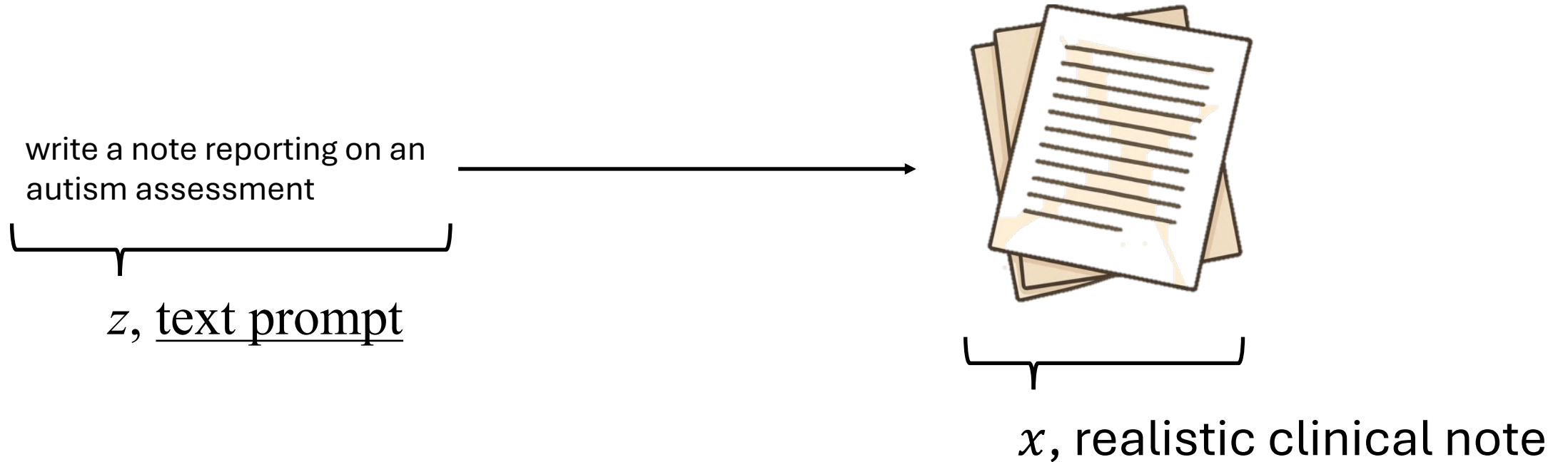


How can we generate **text** from randomness?



End goal: create structure from randomness

How can we generate **text** from a prompt?



End goal: create structure
consistent with the user prompt